

Problem. Verify $f(x) = \frac{3}{2x+1}$ and $g(x) = \frac{3+x}{2x}$ are inverses.

Answer. Write $f(x)$ like this:

$$x = \frac{3}{2y+1}.$$

Now just solve for y :

$$\begin{aligned}x &= \frac{3}{2y+1} \implies (2y+1)x = 3 \\&\implies 2xy + x = 3 \\&\implies 2xy = 3 - x \\&\implies y = \frac{3-x}{2x}.\end{aligned}$$

Then rewrite the last equation like this:

$$f^{-1}(x) = \frac{3-x}{2x}.$$

In your question where you wrote "is this suppose to be negative??", the answer is no. In that step all you do is replace all the y 's with x 's and vice-versa, then solve for y .

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Problem. Graph $f(x) = \frac{(x-1)(x+2)}{(x+1)(x-3)}$.

Answer. You found the vertical asymptotes correctly, all you need to do to find those are set the denominator equal to zero. You also found the horizontal asymptote correctly.

If the degree of the polynomial in the numerator is larger than the degree of the polynomial in the denominator, there are *no horizontal asymptotes*. If the degree of the polynomial in the numerator is *smaller* than the degree of the polynomial in the denominator, then there is a horizontal asymptote at $y = 0$. If the degrees of the polynomials in the numerator and denominator are *equal*, then there is a horizontal asymptote at $y = \frac{a}{b}$, where a is the leading coefficient of the polynomial in the numerator, and b is the leading coefficient of the polynomial in the denominator. Page 300 in the textbook has a table that better explains this.

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Problem. Graph $f(x) = \frac{x^2-2x+1}{x^3-3x^2}$

Answer. The vertical and horizontal asymptotes are correct. Basically for this style of problem you don't need to consider the degree of the entire rational function $f(x)$. You only need to worry about the degree of the numerator and degree of the denominator separately.

In general, the degree of a polynomial is the highest power of x in the equation. As an example, consider the polynomials below:

$$\begin{aligned}f(x) &= x^3 + 2x + 1, \\g(x) &= x^2 + x^{19} + 2x + 1, \\h(x) &= x + 1.\end{aligned}$$

The degree of $f(x)$ is 3, the degree of $g(x)$ is 19, and the degree of $h(x)$ is 1. Notice that the answer is as follows;

$$f(x) = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$